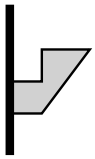
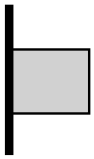
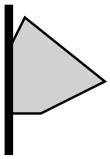
Preview: y2002_q15 (v2)

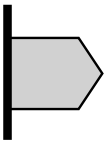
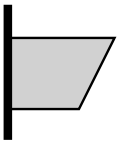
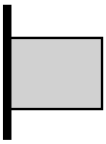
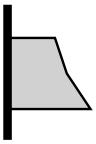
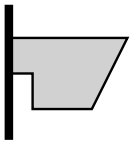
What cannot change when a flag or door moves: the fixed edge stays whole, nothing gets wider

Part A: Each question is worth 3 points

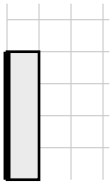
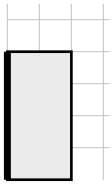
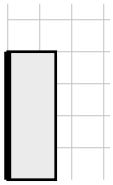
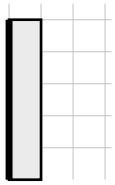
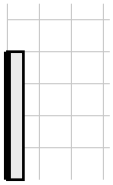
1. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Leo draws the shadow at five different moments.
Which picture cannot be right unless the flag is torn?

A.  ***B.***  ***C.*** ***D.***  ***E.***

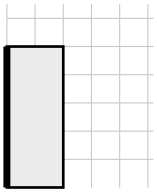
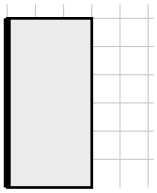
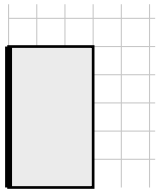
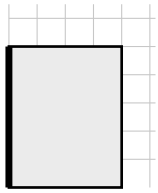
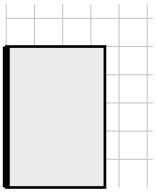
2. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Sara draws the shadow at five different moments.
Which picture cannot be right unless the flag is torn?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

3. A cupboard door is 2 squares wide and 4 squares tall. Its hinges are on the left side (thick line). Tom looks at it straight from the front while it slowly opens, and draws what the door looks like at five moments.
Which picture cannot be right?

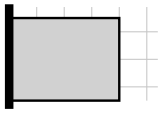
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

4. A cupboard door is 4 squares wide and 5 squares tall. Its hinges are on the left side (thick line). Sara looks at it straight from the front while it slowly opens, and draws what the door looks like at five moments.
Which picture cannot be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

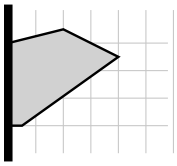
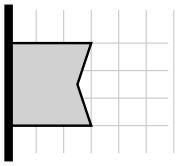
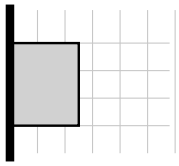
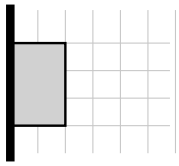
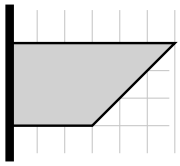
Part B: Each question is worth 4 points

5. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Nina draws the shadow at five different moments. Spread out flat, the flag is 4 squares wide and 3 squares tall (see the picture).

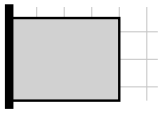


the flag, flat

Which shadow cannot be right?

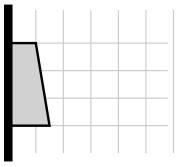
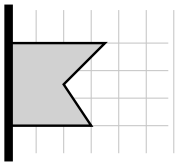
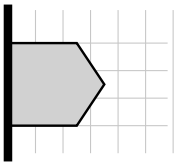
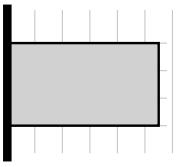
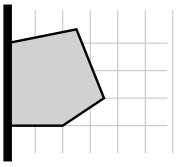
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Tom draws the shadow at five different moments. Spread out flat, the flag is 4 squares wide and 3 squares tall (see the picture).

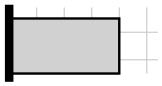


the flag, flat

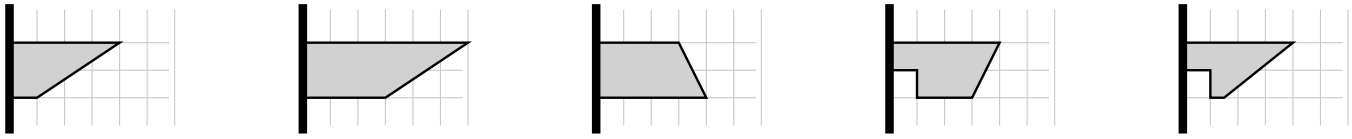
Which shadow cannot be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

7. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Sara draws the shadow at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat

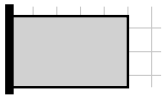


A B C D E

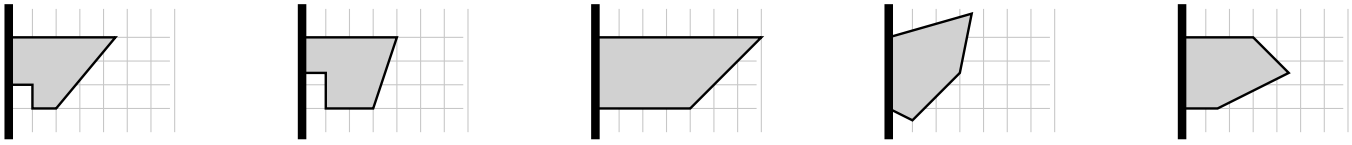
How many of the five drawings can be right?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

8. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Karla draws the shadow at five different moments. Flat, the flag is 5 squares wide and 3 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



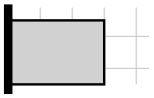
A B C D E

How many of the five drawings can be right?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

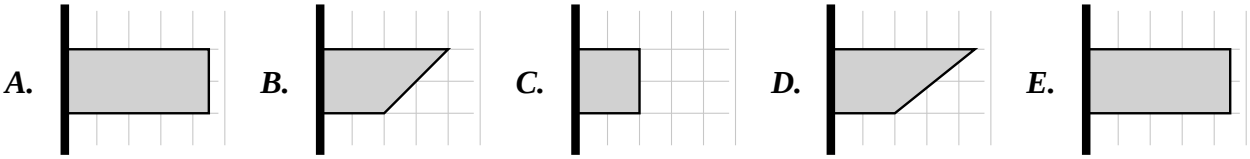
Part C: Each question is worth 5 points

9. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Sara draws the shadow at five different moments. Flat, the flag is 3 squares wide and 2 squares tall (see the picture). Four of the drawings are wrong.

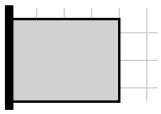


the flag, flat

Which drawing can be right?

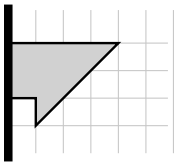


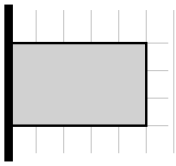
10. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Leo draws the shadow at five different moments. Flat, the flag is 4 squares wide and 3 squares tall (see the picture). Four of the drawings are wrong.

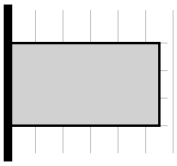


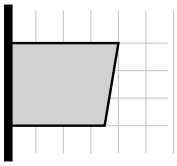
the flag, flat

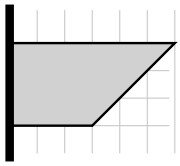
Which drawing can be right?

A. 

B. 

C. 

D. 

E. 

Answer Key

1.	**A**	**2.**	**E**	**3.**	**D**	**4.**	**B**	**5.**	**E**	**6.**	**D**
7.	**C**	**8.**	**C**	**9.**	**C**	**10.**	**D**				

Solutions

1. A — The side sewn to the pole always shows at its full length. In picture A part of the pole is bare, so a piece of the flag is missing. [2002 Q15 · torn_flag]

2. E — The side sewn to the pole always shows at its full length. In picture E part of the pole is bare, so a piece of the flag is missing. [2002 Q15 · torn_flag]

3. D — When a door opens it looks narrower, but it keeps the same height and is never wider than 2 squares. Picture D is wrong: it is taller than the door, but turning never changes the height. [2002 Q15 · door]

4. B — When a door opens it looks narrower, but it keeps the same height and is never wider than 4 squares. Picture B is wrong: it is taller than the door, but turning never changes the height. [2002 Q15 · door]

5. E — Turning or folding can only make the flag look narrower. Picture E reaches more than 4 squares away from the pole, which is wider than the flag. [2002 Q15 · too_wide]

6. D — Turning or folding can only make the flag look narrower. Picture D reaches more than 4 squares away from the pole, which is wider than the flag. [2002 Q15 · too_wide]

7. C (2) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: B, D, E. So 2 can be right. [2002 Q15 · count_possible]

8. C (2) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 5 squares from the pole. Wrong: A, B, C. So 2 can be right. [2002 Q15 · count_possible]

9. C — Check two things: the pole edge must be covered all the way, and nothing may reach more than 3 squares from the pole. Only picture C passes both checks. [2002 Q15 · only_one_right]

10. D — Check two things: the pole edge must be covered all the way, and nothing may reach more than 4 squares from the pole. Only picture D passes both checks. [2002 Q15 · only_one_right]